Phase C0 - Conjecture Synthesis and Proof Search

A transverse rank-one quadratic family with power-saving image sets

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1. One-page result

Exact outcome

The bounded offline run succeeds. Define

$$\boxed{P_*(x,y) = (x+2y)^2 + 3x - y.}$$

There is a fixed constant $\varepsilon > 0$ and a sequence of finite sets $A_j \subset \mathbb{R}$, with $|A_j| \to \infty$, such that

$$\boxed{|P_*(A_j, A_j)| \le |A_j|^{2-\varepsilon}.}$$

Consequently, for every $N_0 \ge 1$, some member of the sequence has $|A_j| \ge N_0$ and satisfies the required inequality. The same set occurs in both inputs.

The proof establishes the following family theorem.

Theorem C1. Let

$$L(x,y) = ax + by, \qquad M(x,y) = cx + ey$$

have integral coefficients, with $\gcd(a,b) = 1$ and

$$ab(ae - bc) \neq 0.$$

Then the fixed polynomial $P = L^2 + M$ has a constant $\varepsilon_P > 0$ and arbitrarily large finite sets $A \subset \mathbb{R}$ satisfying

$$|P(A,A)| \le |A|^{2-\varepsilon_P}.$$

The exponent is not claimed to be uniform over the family. It depends on the fixed coefficients and on one fixed number-field tower and parameter choice.

Status

PROVED WITHIN THE OFFLINE SOURCE SYSTEM. Every load-bearing step is either an inspected statement from a supplied project source, a standard fact stated precisely below, or a complete derivation in this report. The statement has been proved within the current offline source system. Its external novelty and literature status remain unverified.

Mechanism and symbolic exponent

For a squarefree product Q of fixed completely split odd primes, define

$$P_Q = L^2 + QM, \qquad P_Q(Qx, Qy) = Q^2 P(x,y).$$

At each $p \mid Q$, the reduced image is a square set of density $(p+1)/(2p)$. Put

$$\rho = \prod_{p \mid Q} \frac{p+1}{2p}.$$

For the explicit polynomial, a centered Minkowski carrier of radius X gives the exact normalized factor

$$\Theta(X) = D\rho\left(18 + \frac{8Q}{X} + \frac{Q}{X^2}\right).$$

The supplied field construction gives $t \asymp \ell^2$ completely split primes and $\log D = O(\ell \log \ell)$. Since every selected prime is $1 \pmod 4$, $\rho \le (3/5)^t$. Thus one fixes ℓ so that $27D\rho < 1$, and then fixes $X > \max\{Q, \sqrt{D}\}$, giving $\Theta(X) < 1$. One may take

$$\boxed{\varepsilon = \frac{-\log \Theta(X)}{\log(2X+1)} > 0.}$$

Why this is a substantive extension of the supplied example

The supplied benchmark is $(x-y)^2+x+y$. A common affine change applied to both inputs, followed by an affine output change and an optional variable swap, preserves the projective direction of the rank-one quadratic part. The benchmark direction is $[1 : -1]$, whereas the new explicit polynomial has direction $[1 : 2]$. A degree-preserving univariate postcomposition is only an affine output change. Therefore the new polynomial is not obtained from the benchmark by a transport that automatically preserves a Cartesian square.

The advance is not only one new coefficient choice: the proof classifies and establishes compression for the whole transverse square-leading family in Theorem C1.

2. Scope, sources, and epistemic labels

The run remained offline. No browser, web search, arXiv retrieval, remote research service, or network installation was used. All supplied sources were kept read-only. The detailed file-by-file record is in `SOURCE_MANIFEST.md`.

The five Phase B Markdown records were read in full before candidates were selected. The load-bearing field statement is Proposition 3.8 of the supplied unit-distance proof, PDF pages 12-14. The centered carrier correction was checked against frozen Phase A, Section 4.1. The local-to-global framework was checked against Cycle 2, Sections 4-11. Cycle 1 supplied the mandatory warning that collision energy does not upper-bound support.

The labels used below are:

- **PROVED FROM SUPPLIED SOURCES:** an inspected supplied statement is applied with its stated hypotheses;
- **PROVED HERE:** a complete derivation appears in this report;
- **STANDARD FACT:** the exact standard statement used is recorded;
- **ROUTE FAILED:** the tested certificate fails, without refuting the polynomial's compression conjecture;
- **OPEN:** a precise dependency remains unresolved.

No original Pohoata paper was supplied for this run. The Cycle 2 report was therefore used as a supplied-report source, not described as direct primary verification. The final proof does not need arbitrary prescribed-prime compatibility and does not rely on the stronger infinite-split-set statement from the remarks.

3. Candidate generation and selection

Four nonduplicate candidate routes were screened. The detailed cards and eight-component scores are in `CANDIDATE_LEDGER.md`.

ID	Candidate	Score	Decision
C1	$L^2 + M$, with L primitive and M transverse; explicit $(x + 2y)^2 + 3x - y$	39/40	Selected and proved
C2	$(x + 2y)^4 + 3x - y$	28/40	Symmetric-box route failed
C3	$x^2 + xy + y^2 + 3x - y$	29/40	Uniform local route rejected: full image
C4	$x^2 + 3y^2 + x + y$, modulus 3^k	24/40	Exact local theorem; box route failed

Candidate C1 was selected because it is the only route with all four load-bearing features simultaneously proved: sparse reductions at every selected odd prime, common same-input transport, quadratic archimedean scale, and split-prime compatibility without prescribing special congruence classes.

Candidate C2 has stronger fourth-power density but pays a Q^3 coefficient and quartic output cost. Candidate C3 shows that a nondegenerate binary quadratic form is the wrong local residue object. Candidate C4 tests genuine prime-power amplification; its exact density tends to $1/2$, not to zero.

PROJECT CONJECTURE C1 (selection-stage wording). The polynomial $P_* = (x+2y)^2+3x-y$ has a fixed power-saving image sequence on Cartesian squares. External novelty and literature status have not been checked in this offline run. Sections 4-8 prove this conjecture within the offline source system.

4. Algebraic classification and local value sets

4.1 Unique common-scaling quadratic deformation

Write a quadratic polynomial as $P = P_2 + P_1 + P_0$, where the subscripts are homogeneous degrees.

Lemma 4.1 (PROVED HERE). If F_Q has degree at most two and

$$F_Q(Qx, Qy) = Q^2 P(x, y),$$

then necessarily

$$\boxed{F_Q = P_2 + QP_1 + Q^2 P_0.}$$

Proof. Write $F_Q = G_2 + G_1 + G_0$. Then $F_Q(Qx, Qy) = Q^2 G_2 + QG_1 + G_0$. Equality of homogeneous pieces gives $G_2 = P_2$, $G_1 = QP_1$, and $G_0 = Q^2 P_0$. The converse follows by substitution. □

Thus for homogeneous linear forms L, M,

$$P = L^2 + M, qquad P_Q = L^2 + QM, qquad P_Q(Qx, Qy) = Q^2 P.$$

If a constant r is included, the correct deformation is $L^2 + QM + Q^2 r$, not $L^2 + Q(M + r)$.

4.2 Complete binary-quadratic value-set classification

Lemma 4.2 (PROVED HERE). Let q be a homogeneous binary quadratic form over $\mathbb{F}_p$, with p odd.

- rank 0: the value set is $\{0\}$;
- rank 1: the value set has $(p+1)/2$ elements;
- rank 2: the value set is all of $\mathbb{F}_p$.

Proof. A rank-one form is λu^2 in suitable linear coordinates, so its image is a scalar multiple of the squares. A rank-two isotropic form is equivalent to λuv and is visibly surjective. A rank-two anisotropic form is equivalent to $\lambda(u^2 - dv^2)$, with d nonsquare. Identify $u + v\sqrt{d}$ with $\mathbb{F}_{p^2}$. The displayed form is a nonzero scalar multiple of the norm to $\mathbb{F}_p$. Since $\mathbb{F}_{p^2}^{\times}$ is cyclic, the map $z \mapsto z^{p+1}$ has image of order $p-1$, so the norm is surjective. Zero is also represented. □

For an integral quadratic form, rank drops only at the odd primes dividing its discriminant, apart from content primes. Hence a nondegenerate rational quadratic part is locally surjective at all but finitely many odd primes. A primitive rank-one part remains rank one at every odd good prime.

This partially resolves O1 and O2: within the common-scaling quadratic scheme, robust sparse reductions occur exactly when the nonzero homogeneous quadratic part has rank one.

4.3 Exact square images

Lemma 4.3 (PROVED HERE). Let $L = ax + by$ with $\gcd(a, b) = 1$. If $p \mid Q$ is odd, then

$$(L^2 + QM)(\mathbb{F}_p, \mathbb{F}_p) = \{z^2 : z \in \mathbb{F}_p\},$$

and therefore

$$\left|(L^2 + QM)(\mathbb{F}_p, \mathbb{F}_p)\right| = \frac{p+1}{2}.$$

Proof. At least one of a, b is nonzero modulo p, so $L : \mathbb{F}_p^2 \to \mathbb{F}_p$ is surjective. Modulo p, the term QM vanishes. On $\mathbb{F}_p^{\times}$, squaring has kernel $\{1, -1\}$; adding zero gives $(p+1)/2$ values. □

There are no odd exceptional primes for a primitive homogeneous L. At $p = 2$, squaring is the identity and gives no local saving; the field source used below supplies only primes congruent to $1 \pmod 4$.

4.4 Nonspecialness

Lemma 4.4 (PROVED HERE). Let

$$P = (ax + by)^2 + cx + ey + r,$$

with $ab \neq 0$. Then

$$P = g(u(x) + v(y)) \quad \Longleftrightarrow \quad ae - bc = 0,$$

and P is never of the form $g(u(x)v(y))$ with both u, v nonconstant.

Proof. In an additive representation, dependence on both variables makes g, u, v nonconstant. Put $r = \deg g$ and $m = \max\{\deg u, \deg v\}$. The top terms of $u(x)$ and $v(y)$ involve different variables and cannot cancel, so $rm = 2$. If $r = 1$, there is no mixed monomial, contradicting the nonzero xy coefficient. Hence $r = 2, m = 1$, and both inner polynomials are affine linear. Expansion then forces the linear part $cx + ey$ to be parallel to $ax + by$, exactly $ae - bc = 0$. Conversely, parallelism writes P as a univariate quadratic in $ax + by = u(x) + v(y)$.

For a multiplicative representation, write $r = \deg g$, $m = \deg u$, and $n = \deg v$. All are positive and $r(m + n) = 2$, so $r = m = n = 1$. Its homogeneous quadratic part is therefore a scalar multiple of xy, whereas $(ax + by)^2$ has nonzero x^2 and y^2 terms. □

For P_*, $(a, b, c, e) = (1, 2, 3, -1)$, and $ae - bc = -7 \neq 0$. Thus P_* is outside both excluded forms.

5. Number fields and carrier lemmas

5.1 Field supply

Lemma 5.1 (PROVED FROM SUPPLIED SOURCES). For all sufficiently large integers ℓ, put

$$t = \left\lfloor \frac{(\ell-1)^2}{100} \right\rfloor.$$

There exist distinct rational primes $p_1, \dots, p_t$, a totally real field F, and an infinite tower of totally real fields K_j such that $d_j = [K_j : \mathbb{Q}] \to \infty$,

$$\mathrm{rd}(K_j) = \mathrm{rd}(F) =: D, \qquad \log D = O(\ell \log \ell),$$

and every $p_i \equiv 1 \pmod 4$ splits completely in every K_j.

This is Proposition 3.8, properties (P1)-(P4), of the supplied unit-distance proof. The primes are fixed independently of the tower level. Remark 3.9 in that source records the required order of fixed parameters.

For the family theorem, define

$$C_P = 2\left((|a| + |b|)^2 + |c| + |e|\right) + 1.$$

Every selected prime is at least 5, hence its square density is at most $3/5$. Since $t \gg \ell^2$ but $\log D = O(\ell \log \ell)$, one can fix a sufficiently large ℓ for which

$$C_P D \left(\frac{3}{5}\right)^t < 1.$$

After applying Lemma 5.1 at this fixed ℓ, set

$$Q = \prod_{i=1}^{t} p_i, \qquad \rho = \prod_{i=1}^{t} \frac{p_i + 1}{2p_i}.$$

Then Q is a fixed odd squarefree integer, all its prime factors split completely throughout the tower, and

$$\boxed{\theta := C_P D \rho < 1.}$$

No prescribed-prime theorem is used. The dependence of D on ℓ is charged before ℓ is fixed.

5.2 Centered Minkowski carrier

For a totally real field K of degree d, define

$$B_K(X) = \{\alpha \in \mathcal{O}_K : |\sigma_i(\alpha)| \le X \text{ for all } i\}.$$

Lemma 5.2 (PROVED HERE FROM STANDARD FACTS). If $|\Delta_K|^{1/d} \le D$, then

$$\boxed{\left(\frac{X}{\sqrt{D}}\right)^d \le |B_K(X)| \le (2X+1)^d.}$$

Proof. The Minkowski image of $\mathcal{O}_K$ is a lattice of covolume $\sqrt{|\Delta_K|}$. Blichfeldt averaging applied to $[-X/2, X/2)^d$ gives a translate containing at least $X^d/\sqrt{|\Delta_K|}$ lattice points. This is a translated box, not the centered carrier. Fix one of those points and subtract it from all the others. The difference map is injective and lands in $[-X, X]^d$, proving the centered lower bound.

If $\alpha \neq \beta$ are algebraic integers, then $N_{K/\mathbb{Q}}(\alpha - \beta)$ is a nonzero integer. Hence some conjugate difference has absolute value at least 1. The carrier points are therefore 1-separated in sup norm. Packing unit cubes around them inside a cube of side $2X + 1$ proves the upper bound. □

Choose one fixed

$$X > \max\{Q, \sqrt{D}\}.$$

Then $|B_{K_j}(X)| \to \infty$ as $d_j \to \infty$.

5.3 Uniform packing in a residue class

Lemma 5.3 (PROVED HERE). Every class modulo $Q\mathcal{O}_K$ contains at most

$$\left(\frac{2M_0}{Q} + 1\right)^d$$

points of $B_K(M_0)$.

Proof. Distinct congruent points differ by $Q\gamma$ for a nonzero algebraic integer γ. Some conjugate of γ has absolute value at least 1, so the points are Q-separated in sup norm. Disjoint cubes of side Q centered at them lie in a cube of side $2M_0 + Q$. □

This is a worst-case count for each class; no equidistribution is assumed.

6. Complete proof of Theorem C1

Fix $L = ax + by$, $M = cx + ey$, and all parameters as in Section 5. Put

$$P = L^2 + M, \qquad P_Q = L^2 + QM.$$

The algebraic identity and nonspecialness have already been proved. It remains to count the complete support.

Step 1: allowed quotient classes

Complete splitting gives

$$\mathcal{O}_K/p_i\mathcal{O}_K \cong \mathbb{F}_{p_i}^d.$$

The ideals $p_i\mathcal{O}_K$ are pairwise comaximal, so the Chinese remainder theorem gives

$$\mathcal{O}_K/Q\mathcal{O}_K \cong \prod_{i=1}^{t} \mathbb{F}_{p_i}^d.$$

At each of the d coordinates for p_i, Lemma 4.3 permits exactly $(p_i + 1)/2$ values. Hence every output reduction lies in a set of at most

$$\prod_{i=1}^{t}\left(\frac{p_i+1}{2}\right)^d = \boxed{(\rho Q)^d}$$

classes modulo $Q\mathcal{O}_K$. This is a Cartesian-product count in the quotient ring, not probabilistic independence. Surjectivity of the carrier onto the quotient is neither asserted nor needed.

Step 2: all-conjugate output radius

For $\alpha, \beta \in B_K(X)$, every real embedding satisfies

$$|L(\alpha,\beta)| \le (|a|+|b|)X, \qquad |M(\alpha,\beta)| \le (|c|+|e|)X.$$

Since $X > Q$,

$$|\sigma(P_Q(\alpha,\beta))| \le \left((|a|+|b|)^2 + |c| + |e|\right) X^2 =: M_0.$$

Thus $P_Q(B_K(X), B_K(X)) \subset B_K(M_0)$.

For the explicit polynomial, the sharper exact radius is

$$M_0 = 9X^2 + 4QX,$$

which gives

$$\Theta(X) = D\rho\left(18 + \frac{8Q}{X} + \frac{Q}{X^2}\right) < 27D\rho < 1.$$

Step 3: full-support multiplication

By Lemma 5.3, each allowed class contributes at most $(2M_0/Q+1)^d$ outputs. Therefore

$$\begin{aligned}|P_Q(B_K(X), B_K(X))| &\le (\rho Q)^d\left(\frac{2M_0}{Q}+1\right)^d \\ &= [\rho(2M_0+Q)]^d.\end{aligned}$$

Put $n = |B_K(X)|$. The carrier lower bound gives $X^{2d} \le D^d n^2$. Since $2M_0 + Q \le C_P X^2$,

$$\boxed{|P_Q(B_K(X), B_K(X))| \le \theta^d n^2, \qquad \theta = C_P D\rho < 1.}$$

This count covers every output. Collision energy is not used.

Step 4: real embedding and fixed-polynomial transport

Choose one real field embedding $\sigma_j : K_j \hookrightarrow \mathbb{R}$ and define

$$A_j = Q^{-1}\sigma_j(B_{K_j}(X)).$$

A field embedding is injective, and evaluation of an integral polynomial commutes with it. The same scalar map is used in both input coordinates. Thus

$$|A_j| = |B_{K_j}(X)| =: n_j$$

and, from $P_Q(Qx, Qy) = Q^2 P(x, y)$,

$$|P(A_j, A_j)| = |P_Q(B_{K_j}(X), B_{K_j}(X))| \le \theta^{d_j} n_j^2.$$

There are no input or output fibers to charge.

Step 5: exponential-to-power conversion

Define

$$\varepsilon_P = \frac{-\log \theta}{\log(2X+1)} > 0.$$

The direction of the conversion uses the carrier **upper** bound:

$$n_j \le (2X+1)^{d_j} \quad \Longrightarrow \quad n_j^{\varepsilon_P} \le \theta^{-d_j}.$$

Hence

$$|P(A_j, A_j)| \le n_j^{2-\varepsilon_P}.$$

The carrier lower bound and $X > \sqrt{D}$ show $n_j \to \infty$. Therefore the sets meet every threshold N_0 along the sequence. This completes the proof of Theorem C1 and its explicit corollary. □

7. Parameter ledger

Parameter	Fixed when	Depends on	Gain or cost
P, L, M, C_P	first	explicit coefficient choice	fixes the theorem; C_P is archimedean cost
ℓ	second	C_P and source thresholds	makes quadratic split-prime gain dominate $\log D$
t, D	with the source tower	ℓ	t supplies savings; D costs input density
$p_1, \dots, p_t$	after ℓ	Proposition 3.8 construction	split square restrictions
Q	after the primes	their product	fixed deformation coefficient and modulus
ρ	after the primes	exact square densities	local gain $-\log \rho$
X	after Q, D	chosen greater than both Q and $\sqrt{D}$	carrier and output scale
d_j	varies last	tower level	amplifies the already positive fixed rate

All finite choices are fixed before $d_j \to \infty$. Increasing the number of primes would require recomputing D, Q, ρ, X; the proof instead fixes one sufficiently large ℓ once.

8. Proof dependency diagram

Stage	Claim and status	Output passed forward
1	Proposition 3.8, **SUPPLIED SOURCE**	Totally real tower, $d_j \to \infty$, split primes, D
2	Quadratic rate comparison, **PROVED HERE**	Fixed Q, ρ with $C_P D \rho < 1$
3	Deformation identity, **PROVED HERE**	$P_Q(Qx, Qy) = Q^2 P(x, y)$
4	Square value sets and CRT, **PROVED HERE + STANDARD CRT**	At most $(\rho Q)^d$ output classes
5	Centered carrier and output radius, **PROVED HERE**	Large input and one bounded all-conjugate region
6	Worst-class packing, **PROVED HERE**	Full support $\le [\rho(2M_0 + Q)]^d$
7	Signed rate, **PROVED HERE**	$\lvert P_Q(B, B)\rvert \le \theta^d \lvert B\rvert^2$, $\theta < 1$
8	Common scalar transport, **PROVED HERE**	Same real set in both inputs
9	Upper-size exponent conversion, **PROVED HERE**	$\lvert P(A, A)\rvert \le \lvert A\rvert^{2-\varepsilon_P}$

There is no assumed or open arrow in the proof of Theorem C1.

9. Adversarial audit

The independent referee report is reproduced separately in `REFEREE_REPORT.md`. The referee independently reconstructed the algebra, field supply, full-support count, transport, and exponent conversion, and returned the verdict **PROVED WITHIN THE OFFLINE SOURCE SYSTEM**.

The serious objections and dispositions were:

1. **Overbroad nonspecial family wording.** Primitive L and transverse M are insufficient if L uses only one variable: $x^2 + y$ is additive-special. **Resolved:** Theorem C1 assumes $ab \neq 0$, as well as transversality. The compression mechanism itself works more broadly, but the nonspecial claim is made only under the corrected hypotheses.
2. **Compressed degree argument.** A nonspecialness proof must eliminate all degree factorizations. **Resolved:** Lemma 4.4 treats the linear and quadratic additive cases and the exact multiplicative total-degree case.
3. **Prescribed-prime overreach.** A generic tower statement would be too weak. **Resolved:** Lemma 5.1 cites Proposition 3.8 directly and uses only the primes produced by that construction.
4. **Rate notation.** The exact factor $\Theta(X)$ and conservative family factor $\theta = C_P D \rho$ must not be conflated. **Resolved:** They have distinct names and each exponent is defined from the factor used.
5. **Extension scope.** The direction invariant does not rule out arbitrary two-variable affine equivalence. **Resolved:** The claim is restricted to common affine input maps, affine output maps, and variable swap—the operations relevant to automatic preservation of a Cartesian square.

6. **F1-F10 and Section 8 checks.** Every check passed: no collision-energy inference, no hidden fibers, no equidistribution, correct complete splitting and CRT, centered carrier correction, all-conjugate control, fixed parameters, and correct upper-bound exponent conversion.

No repair changed the primary polynomial, construction, rate inequality, or final quantifiers.

10. Failure and mutation history

C2: quartic symmetric-box route

For

$$P^{(4)} = (x+2y)^4 + 3x - y, \qquad P_Q^{(4)} = (x+2y)^4 + Q^3(3x-y),$$

the identity $P_Q^{(4)}(Qx, Qy) = Q^4 P^{(4)}(x,y)$ holds. At each supplied split prime, the density of fourth powers is $(p+3)/(4p)$. The exact symmetric-box radius is

$$M_4 = 81X^4 + 4Q^3X,$$

so

$$\Theta_4(X) = D\rho_4 \left(162X^2 + \frac{8Q^3}{X} + \frac{Q}{X^2}\right).$$

Writing $z = X/Q$, either $z \geq 1/4$ and $162z^2 > 10$, or $z < 1/4$ and $8/z > 32$. Therefore the first two terms exceed $10Q^2$. Also

$$\rho_4 Q^2 = \prod_{p \mid Q} \frac{p(p+3)}{4} \geq 10^t.$$

Hence $\Theta_4(X) > 1$ for every $X > 0$. **ROUTE FAILED.** No conclusion is drawn about other carriers or about the quartic conjecture itself.

C3: full-rank quadratic local image

For $q = x^2 + xy + y^2$, the discriminant is -3. Every supplied split prime is $1 \pmod 4$, hence is not 3. Lemma 4.2 makes q surjective at all selected primes. The local density is 1, so the route has no local gain. **LOCAL ROUTE REJECTED; CONJECTURE NOT REFUTED.**

C4: prime-power test

For every $k \geq 1$, exact valuation analysis gives

$$\left|\{x^2 + 3y^2 \pmod{3^k} : x, y \in \mathbb{Z}/3^k\mathbb{Z}\}\right| = \frac{3^k+1}{2}.$$

Indeed, the valuations $2v_3(x)$ and $1 + 2v_3(y)$ have opposite parity and cannot cancel; every represented nonzero unit part is $1 \pmod 3$, and each such unit is a square modulo every 3^j. Summing over possible valuations gives the displayed formula. Its density tends to $1/2$, and the exact box

ledger remains greater than $4D$. The supplied source also does not guarantee complete splitting of the prescribed prime 3. **ROUTE FAILED; CONJECTURE NOT REFUTED.**

No mutation of C1 was needed after the integrated proof and audit. The run used one successful serious iteration and three bounded screening iterations, within the six-iteration limit.

11. Mechanism progress relative to O1-O7

Open lemma	Phase C0 progress
O1: sparse local value-set criterion	**Resolved for all homogeneous binary quadratics over odd fields.** Rank one gives a square set; rank two is surjective. Still open for arbitrary bivariate polynomials.
O2: removable arithmetic deformation	**Resolved for common-scaling quadratics and proved for the transverse rank-one family.** The unique degree weighting is $P_2 + QP_1 + Q^2P_0$. General deformations remain open.
O3: higher-degree threshold	**Narrowed negatively.** The quartic power mutation fails the symmetric-box certificate for every scale; generic height screens give the same obstruction more broadly.
O4: prescribed split primes	**Avoided, unchanged.** C1 is sparse at every odd selected prime, so Proposition 3.8's own primes suffice.
O5: prime powers	**Narrowed by one exact theorem.** The $x^2 + 3y^2$ image modulo 3^k has $(3^k + 1)/2$ values, insufficient here. General lifting remains open.
O6: rank-two correlated sums	**Unchanged and unnecessary for C1.**
O7: carriers beyond boxes	**Unchanged.** It is the natural escape route for higher degree after C2.

The main advance beyond the supplied example is the proof of an entire same-set family and the local classification explaining why rank-one quadratic leading parts are the robust quadratic source of sparse reductions.

12. Prioritized next bottleneck

The next prioritized lemma is a higher-degree carrier statement, not another unstructured polynomial search.

> **Proposed O7/O3 carrier lemma.** Fix an even $m > 2$, primitive two-variable integral L, and transverse integral linear M. For the common-scaling deformation
>
> $$F_Q = L^m + Q^{m-1}M, \qquad F_Q(Qx, Qy) = Q^m(L^m + M),$$
>
> construct, or rule out, carriers $C_d \subset \mathcal{O}_{K_d}$ in a supplied split-prime tower for which:
>
> $$e^{ad-o(d)} \le |C_d| \le e^{Bd+o(d)},$$
>
> every allowed output class has at most $e^{ud+o(d)}$ points in the exact output region, and
>
> $$2a - (\log R + u) - \log \rho_m > 0,$$

with every coefficient, modulus, and fiber cost included.

This lemma is narrower than the target theorem: it asks for a specific uniform class-packing certificate for a fixed higher-power deformation. The quartic calculation proves that the symmetric Minkowski box cannot supply it, so any positive answer must introduce genuinely new carrier geometry.

The full remaining-gap list is: arbitrary sparse-value criteria beyond quadratics, general removable deformations, general prime-power lifting, prescribed split-prime compatibility, rank-two correlated sums, and non-box carriers.

13. Final status

Exact classification: PROVED WITHIN THE OFFLINE SOURCE SYSTEM.

Strongest established conclusion: every fixed transverse square-leading integral polynomial in Theorem C1, in particular

$$(x + 2y)^2 + 3x - y,$$

has arbitrarily large same-input finite real sets with a fixed power-saving image bound.

Remaining mathematical bottleneck: construct a higher-degree carrier or uniform class-packing theorem that beats the explicit degree and deformation cost; the symmetric-box mechanism provably does not do so for the tested quartic family.

The statement has been proved within the current offline source system. Its external novelty and literature status remain unverified. Phase C0 stops here.